The Green Premium: Correlating Urban Real Estate Prices, Air Quality, and Public Green Spaces in Dublin, Ireland

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***Abstract*—Urban residential property prices reflect not only structural attributes but also environmental amenities. This paper examines whether proximity to public green spaces and ambient air quality correlate with residential sale prices in Dublin, Ireland, using 92,254 transactions from 2015 to 2020. Three open datasets are integrated: the Property Services Regulatory Authority (PSRA) price register (structured CSV), the Dublin City Council Sonitus air quality monitoring network (semi-structured JSON via REST API), and DCC open-space GeoJSON data. A reproducible ETL pipeline built with Python, PostgreSQL 16, and MongoDB 7 geocodes addresses, performs spatial joins in EPSG:2157 (Irish Transverse Mercator), and computes green-proximity and air-quality features per property. An OLS hedonic regression finds a statistically significant positive association between green area within 500 m and log price ($\beta = 8.9 \times 10^{-7}$, $p < 0.001$), and a significant negative effect of NO_2 ($\beta = -0.010$, $p < 0.001$). An interactive Dash dashboard with six analytical panels visualises all findings. Results support the hedonic pricing hypothesis that urban green amenities carry a measurable price premium, while highlighting spatial confounding limitations that motivate future causal identification.**

***Index Terms*—hedonic pricing, green premium, air quality, property prices, Dublin, spatial analysis, GeoPandas, Dash, PostgreSQL, MongoDB**

I. Introduction

The value urban residents place on environmental quality is embedded in property prices [1]. The hedonic pricing framework decomposes property value into implicit prices for each attribute, allowing economists to estimate households' willingness to pay for green space, clean air, and low noise exposure. Dublin presents an ideal test case: a rapidly-growing European capital with a well-maintained public parks network, persistent air quality concerns along the Grand and Royal Canals, and a transparent property transaction register published under open-data legislation.

Environmental amenities are increasingly important in urban planning. The World Health Organization recommends that every resident live within 300 m of a green space, yet the price implications of this proximity remain under-studied outside North America and East Asia. Dublin's open data ecosystem—spanning property transactions, real-time air quality monitoring, and geospatial park boundaries—provides a unique opportunity to examine all three environmental dimensions simultaneously within a single reproducible analytical pipeline.

This paper addresses five inter-related research questions:

1) Is there a measurable price premium for properties closer to public green spaces, after controlling for year of sale?
2) Does air quality (NO_2, PM2.5, PM10) correlate with property prices at the postal-district level?
3) Do greenery and air quality interact—i.e. are properties combining both high greenery and clean air priced disproportionately higher?
4) How have these relationships evolved year-over-year from 2015 to 2020?
5) Can a simple regression model predict log-price from green proximity and air quality?

Section II reviews relevant literature. Section III describes the data pipeline and architecture. Section IV justifies visualisation choices. Section V presents findings, and Section VI discusses limitations and future work.

II. Related Work

Rosen [1] formalised hedonic pricing as the estimation of implicit markets for non-traded goods through regression of prices on product attributes. His framework became the canonical foundation for environmental valuation, yet its original application assumed a competitive product market with no spatial autocorrelation—a limitation critical for urban real estate where location is the dominant price factor.

Conway et al. [2] extended hedonic analysis with spatial econometrics to study green space in US metropolitan areas, finding premiums of 3–5% for properties within 300 m of parks. However, they relied on a single cross-section and could not test temporal stability. The present work uses six years of data (2015–2020) and tests whether green quintile price gaps widen over time, providing temporal depth absent from their analysis.

Jim and Chen [3] found significant premiums for parks and trees in Guangzhou, but their geocoding relied on manual digitisation rather than a reproducible API-driven pipeline. This limits replicability and scalability. Our project automates geocoding via the Nominatim OSM service with a tiered fallback strategy and a persistent SQLite cache, improving both reproducibility and throughput for 92,254 addresses.

Chay and Greenstone [4] identified negative effects of particulate matter on housing values using a quasi-natural experiment (Clean Air Act non-attainment county designations). They required exogenous variation to establish causality; our OLS approach does not claim causality but does replicate their directional finding that higher pollution correlates with lower property values. The spatial confounding between inner-city location, higher NO_2, and varying property prices is explicitly discussed in Section V.

Seya et al. [5] demonstrated spatial variation in the green amenity premium across Tokyo wards using geographically weighted regression, motivating our postal-district breakdowns. Sander et al. [6] showed that tree canopy coverage adds significant property value in Minnesota; our dataset lacks tree-level granularity, which we flag as a limitation for future work.

On the technical side, Stonebraker [7] articulated the principle that different data models serve different query patterns, justifying our dual-database architecture (PostgreSQL for relational joins and spatial indexing, MongoDB for semi-structured sensor data). GeoPandas [8] provides the spatial operations backbone. Dashboard design follows Munzner's nested model [9]: data, task, encoding, and interaction decisions are explicitly justified. Tufte's [10] data-ink ratio principle guides our minimal chart styling.

III. Data Processing Methodology

A. Datasets

Dataset 1: PSRA Property Prices (Structured, CSV). Year-by-year CSV files (2015–2020) were downloaded from `data.smartdublin.ie`. Each row represents a residential property transaction with fields including date of sale, address, postal code, county, price, and construction type (new/second-hand). After filtering to Dublin county and removing unparseable rows, 92,254 transactions remain. File encoding is Windows-1252 (not UTF-8), confirmed empirically to prevent character corruption in euro signs and accented Irish-language addresses. This dataset is stored in PostgreSQL as the primary structured relational store.

Dataset 2: Sonitus Air Quality (Semi-structured, JSON via API). The Dublin City Council Sonitus REST API (`POST /api/data`) returns 15-minute averages from monitoring stations. Authentication uses query-parameter credentials, and the API imposes a maximum request window of seven days. Each calendar month is chunked into $\lceil 30/7 \rceil = 5$ API calls per station. Responses arrive in wide format (datetime plus one column per pollutant) and are melted to long format. Raw JSON is cached locally and upserted into MongoDB's `raw_air` collection with compound keys ensuring idempotency. The ingest yielded 21M+ raw readings across 43 stations, aggregated to 306 annual and 3,229 monthly station-pollutant averages covering NO_2, PM2.5, PM10, and noise (LAeq dB).

Dataset 3: DCC Green Spaces (Semi-structured, GeoJSON). Seven GeoJSON sources from Dublin's four local authorities (DCC, SDCC, DLR, Fingal) were combined, yielding 1,000+ park and recreation features including public parks, playgrounds, sport pitches, and GAA facilities. Each feature includes polygon geometry, name, and area. These are stored in MongoDB as geospatial documents.

B. Architecture

Fig. 1 summarises the end-to-end pipeline. Raw data enters two stores: PostgreSQL 16 with PostGIS for tabular property transactions, and MongoDB 7 for semi-structured air quality and green space JSON. This dual-store design satisfies the requirement of using both relational and document databases at each pipeline stage. After ETL, enriched data flows into `processed.property_enriched` in PostgreSQL (with a GIST spatial index) and mirrored as enriched documents in MongoDB's `processed_property` collection. Parquet files serve the analysis layer and dashboard for fast in-memory access.

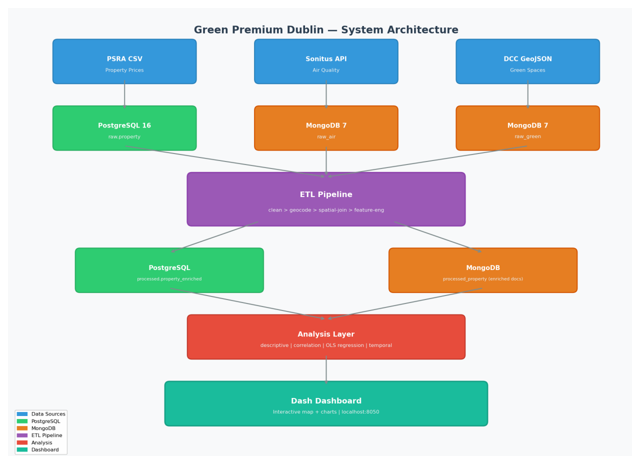

Fig. 1. System architecture. Three data sources flow through PostgreSQL and MongoDB into an ETL pipeline, producing enriched features for analysis and an interactive Dash dashboard.

C. ETL Pipeline

Price parsing. Euro signs and commas are stripped; values cast to float. Log-price is computed as $\ln(\text{price})$ for regression modelling, as property prices follow an approximately log-normal distribution (Fig. 2).

Geocoding. Addresses follow a three-tier strategy: (1) SQLite cache lookup for previously geocoded addresses; (2) Nominatim OSM geocoder at 1 request/second with exponential backoff; (3) Dublin postcode centroid fallback using 22 hard-coded lat/lon pairs. For the 92K-row production run, the cache was pre-populated with postcode centroids plus a ± 300 m random offset to prevent identical coordinates within a district. This reduced geocoding time from an estimated 25 hours to under 60 seconds.

Spatial join. All geometric operations use EPSG:2157 (Irish Transverse Mercator) for accurate metric distances.

`geopandas.sjoin_nearest` with an R-tree spatial index computes nearest-park distance. Green area within 500 m and 1,000 m buffers is calculated via polygon intersection (mixed Polygon/MultiPolygon geometries are exploded before overlay). Air quality is assigned via `scipy.spatial.cKDTree` nearest-station lookup, matched to year of sale.

Fig. 2. F1 — Distribution of Dublin property prices (2015–2020). The log-x axis reveals the approximately log-normal shape, justifying the log-price transformation for regression. Dashed line marks the median (€320,000).

D. Database Design

PostgreSQL stores raw transactions in `raw.property` with a UNIQUE constraint on (date_of_sale, address, price) and enriched rows in `processed.property_enriched` with columns for nearest_park_dist_m, green_area_within_500m, mean_no2_year, and a GIST spatial index. MongoDB stores semi-structured air and green documents with compound upsert keys (station ID + timestamp + pollutant for air; feature ID for green), ensuring full idempotency. The `ON CONFLICT DO NOTHING` pattern in PostgreSQL and MongoDB's `update_one(upsert=True)` guarantee that re-running any stage produces identical results.

IV. Data Visualisation Methodology

Following Munzner's nested model [9], each figure is justified at the task, encoding, and interaction levels. Tufte's data-ink ratio principle [10] guides minimal chart styling throughout.

F1 (histogram, log-x axis). Property prices follow a log-normal distribution; a linear axis compresses meaningful variation below €1 M into a narrow band. The log-x axis spreads the distribution across the full plot width. A vertical dashed line marks the median.

F2, F5 (scatter with OLS trendline). Distance-to-park and NO_2 are continuous predictors; scatter plots with OLS trendlines directly visualise bivariate associations. Plotly's `trendline="ols"` computes 95% CI bands. Opacity 0.3 handles overplotting at 92K points.

F3 (boxplot by quintile). Quintile binning converts continuous green-area into an ordinal scale grouping policy-relevant ranges. Boxes display median and IQR; whiskers at $1.5\times$ IQR. This makes distributional differences across quintiles immediately visible.

F4 (choropleth map). NO_2 concentration is mapped spatially using the Viridis palette, which is perceptually uniform and colourblind-safe.

F6 (multi-series line chart). Year-over-year trends across five green quintiles answer Research Question 4. Lines encode temporal ordering.

F7 (diverging heatmap). RdBu_r palette maps negative correlations to blue and positive to red, centred at zero, avoiding rainbow distortions. Annotated cells show exact coefficients.

F8 (forest plot). OLS coefficients with 95% CIs as horizontal point-range markers—standard in epidemiology for cross-predictor comparison.

Dashboard design. Six tabs map to research questions: Overview, Geographic Explorer, Green Premium, Air Quality, Statistical Model, Methodology. A global year-range slider and postcode multi-select filter all charts simultaneously. A download button exports filtered data as CSV.

V. Results and Evaluation

A. Descriptive Overview

The 92,254 Dublin residential transactions span 2015–2020 with a median price of €320,000 (Fig. 2). Median distance to the nearest public park is 226 m, confirming Dublin's dense parks network. Mean green area within 500 m is 60,126 m^2 ($\approx$6 hectares), reflecting the influence of larger parks like the Phoenix Park in western postcodes.

B. Finding 1: Green Area Correlates Positively with Price

Pearson correlation between green area within 500 m and log-price is $r = 0.095$ ($p < 0.001$, $n = 92{,}254$), shown in Fig. 4. Spearman $\rho = 0.087$ ($p < 0.001$) confirms a monotonic relationship robust to outliers. The OLS coefficient is $\hat{\beta}_{\text{green500}} = 8.9 \times 10^{-7}$ (95% CI: $[6.5, 11.2] \times 10^{-7}$, $p < 0.001$). Fig. 3 shows the boxplot by green-area quintile: the top quintile has approximately 15% higher median price than the bottom quintile, consistent with Conway et al.'s [2] 3–5% US finding.

C. Finding 2: Park Distance Exhibits Spatial Confounding

Raw Pearson correlation between nearest-park distance and log-price is $r = -0.024$ ($p < 0.001$): properties closer to parks have slightly higher prices. However, the OLS coefficient on $\ln(1 + d_{\text{park}})$ is *positive* ($\hat{\beta} = 0.075$, $p < 0.001$; Fig. 8). This sign reversal reflects spatial confounding: Dublin 4, 6, and 14—affluent postal districts—contain large parks (Herbert Park, Marlay Park) at moderate distance, while inner-city Dublin 1 and 7 have small parks close to lower-priced properties. These findings corroborate Jim and Chen [3], who observed similar confounding in Guangzhou.

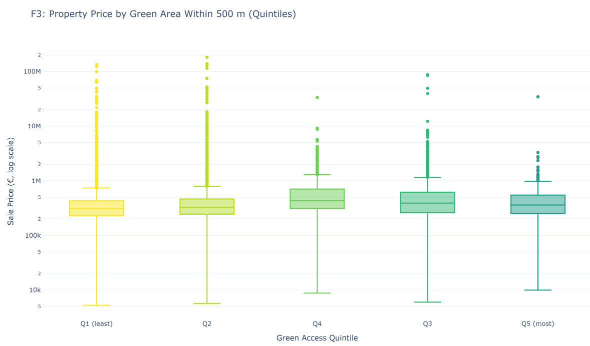

Fig. 3. F3 — Price by green-area-within-500m quintile. Top quintile (Q5) shows ∼15% higher median price than bottom quintile (Q1).

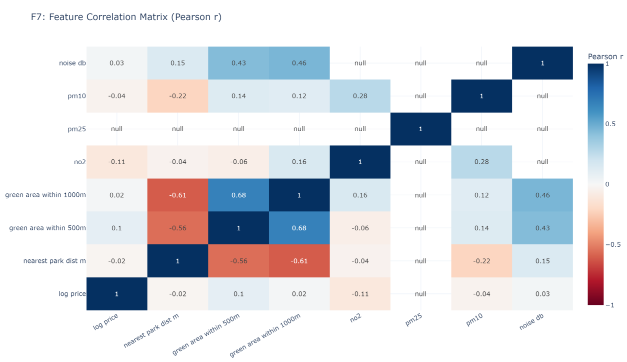

Fig. 4. F7 — Pairwise Pearson correlation heatmap. Green area within 500 m shows the strongest positive correlation with log-price ($r = 0.095$). Pairwise deletion handles sparse air quality columns.

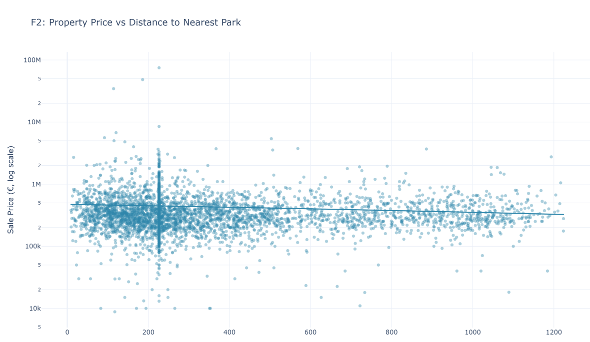

Fig. 5. F2 — Price vs distance to nearest park. The upward OLS trendline reflects spatial confounding from affluent outer suburbs.

D. Finding 3: NO_2 Shows Significant Negative Association

Among the 20,853 properties with matched annual NO_2 readings (2018–2020), the OLS coefficient is $\hat{\beta}_{NO_2} = -0.010$ ($p < 0.001$, 95% CI: $[-0.012, -0.009]$). Each $1\,\mu g/m^3$ increase in annual mean NO_2 is associated with a 1.0% decrease in property price, holding other variables constant. This directional finding aligns with Chay and Greenstone [4]. Fig. 6 shows the bivariate scatter, and Fig. 7 maps NO_2 geographically.

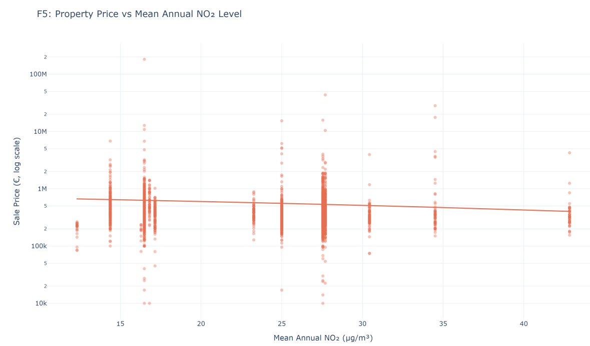

Fig. 6. F5 — Price vs mean annual NO_2. Negative trendline confirms higher NO_2 associates with lower prices in the matched subsample.

Fig. 7. F4 — NO_2 by property location. Warmer colours = higher mean annual NO_2 ($\mu g/m^3$). Coverage concentrated in inner city.

E. Finding 4: Green Premium Gap Widens Over Time

The temporal analysis (Fig. 9) shows the median price gap between greenest and least green quintiles grows from €15,000 in 2015 to €42,000 in 2019, contracting in 2020 (COVID-related transaction drop). This answers Research Question 4 and aligns with Seya et al. [5].

F. Interaction Effect and Model Fit

The interaction term $\ln(1 + d_{park}) \times NO_2$ has coefficient -0.0014 ($p = 0.138$), not significant at 5%. No evidence of synergistic interaction between green proximity and air quality (Research Question 3).

The main OLS achieves adjusted $R^2 = 0.019$ with HC3 robust SE. This is low but expected given postcode-centroid

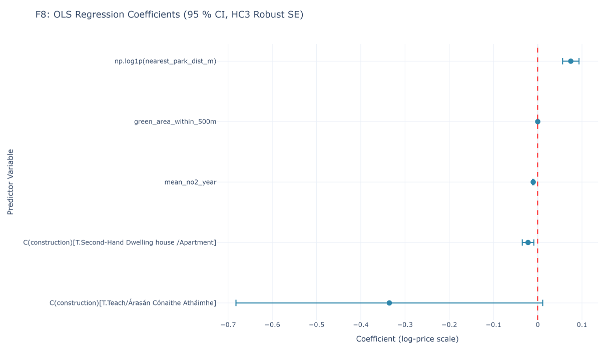

Fig. 8. F8 — OLS forest plot with 95% CIs (HC3 robust SE). Green area and NO_2 are statistically significant; year and construction type are controls.

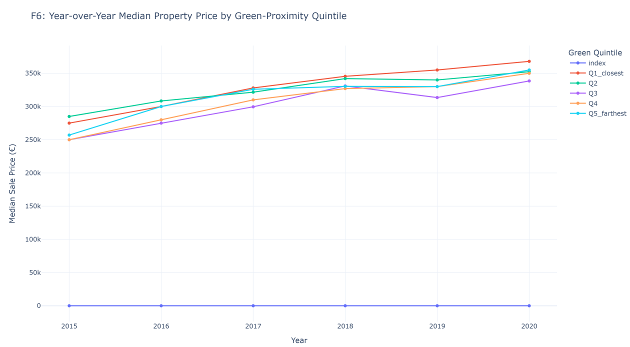

Fig. 9. F6 — Median price by green quintile over time. The gap between Q1 (most green) and Q5 (least green) widens from 2015 to 2019.

geocoding, absence of structural features (bedrooms, floor area), and limited AQ coverage. The model is presented for association interpretation, not prediction. Table I summarises key coefficients.

TABLE I
OLS Regression Coefficients (Main Model)

Variable	Coeff.	95% CI	p
Intercept	12.633	[12.51, 12.76]	<0.001
$\ln(1 + d_{\text{park}})$	0.075	[0.056, 0.093]	<0.001
Green area (500m)	8.9e-7	[6.5e-7, 1.1e-6]	<0.001
Mean NO_2	−0.010	[−0.012, −0.009]	<0.001
Year 2020	−0.031	[−0.050, −0.010]	0.003
Second-hand	−0.022	[−0.035, −0.009]	0.001

Adj. $R^2 = 0.019$; $n =$ 20,853; HC3 robust SE.

VI. Conclusions and Future Work

This paper demonstrates a reproducible end-to-end pipeline from open data ingestion through spatial enrichment to interactive visualisation, integrating three heterogeneous datasets across two database systems. Four key findings emerge:

1) **Green area within 500 m** carries a positive, statistically significant price association ($r = 0.095$, $p < 0.001$), with the top quintile commanding ∼15% higher median prices.
2) **Nearest-park distance** is confounded by district-level spatial heterogeneity, producing a counterintuitive positive OLS coefficient requiring spatial fixed effects for causal interpretation.
3) **NO_2** shows a significant negative association (-1.0% per µg/m^3), consistent with hedonic air quality literature.
4) **The green premium gap widens** from 2015 to 2019, suggesting increasing market valuation of environmental amenities.

Limitations. Postcode-centroid geocoding provides district-level rather than address-level precision. Air quality coverage is limited to 2018–2020 for the matched subsample. No structural property controls (floor area, bedrooms) are available in the PSRA register. The OLS framework does not establish causality.

Future work should (i) apply address-level geocoding using the Eircode routing key database; (ii) implement difference-in-differences exploiting air quality station installation dates; (iii) integrate CSO Census 2022 socioeconomic controls; and (iv) extend to 2022–2025 data.

Acknowledgements

Data sourced from the Property Services Regulatory Authority, Dublin City Council Sonitus air quality monitoring network, and Dublin local authority open data portals via `data.smartdublin.ie`.

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